- The symbol looks like an elongated S [$\int$], which stands for "summation" is the sign of integration.
- $\int f(x)\,dx$ is read as integral of $f(x)$ with respect to x.
- $f(x)$ in $\int f(x)\,dx$ [i.e. the function to be integrated] is called as integrand.
- x in $\int f(x)\,dx$ is the variable of integration.
- The term integration means the process of finding the integral.
- The term constant of integration means any real number c, considered as a constant function.

Definition 2.2

The process of determining an integral of a given function is defined as integration of a function.

2.1.2 Two important properties of Integral Calculus

(i) if k is any constant, then $\int k\,f(x)\,dx = k\int f(x)\,dx$

(ii) if $f(x)$ and $g(x)$ are any two functions, then $\int[f(x)\pm g(x)]\,dx = \int f(x)\,dx \pm \int g(x)\,dx$

Methods of Integration

The following are the four principal methods of integration.

(i) Integration by decomposition.
(ii) Integration by parts.
(iii) Integration by substitution.
(iv) Integration by successive reduction.

Remember !

Integrate a function $f(x)$ means, finding a function $F(x)$ such that $\frac{d}{dx}[F(x)+c] = f(x)$

Note

Here, we discuss only the first three methods of integration, because the method of integration by successive reduction is beyond the scope of the syllabus.

2.1.3 Integration by decomposition

Apply this method, whenever the integrand is impossible to integrate directly, but it can be integrated after decomposing into a sum or difference of integrands whose individual integrals are already known.

Note

The integrals which are directly obtained from their corresponding derivatives are known as the standard results of integration.

Some standard results of integration

Type: I

(i) $\int x^n\,dx = \frac{x^{n+1}}{n+1} + c,\ n \neq -1$

(ii) $\int (ax+b)^n\,dx = \frac{(ax+b)^{n+1}}{a(n+1)} + c,\ n \neq -1$

DO YOU KNOW?

$y = \int f(x)\,dx = F(x) + c$ denotes family of curves having parallel tangents at $x = k$

Example 2.1

Evaluate $\int \frac{ax^2+bx+c}{\sqrt{x}}\,dx$

Solution:

$$\int \frac{ax^2+bx+c}{\sqrt{x}}\,dx$$

$$= \int \left(ax^{\frac{3}{2}} + bx^{\frac{1}{2}} + cx^{-\frac{1}{2}} \right) dx$$

$$= a\int x^{\frac{3}{2}}\,dx + b\int x^{\frac{1}{2}}\,dx + c\int x^{-\frac{1}{2}}\,dx$$

$$= \frac{2ax^{\frac{5}{2}}}{5} + \frac{2bx^{\frac{3}{2}}}{3} + 2cx^{\frac{1}{2}} + k$$

DO YOU KNOW?

$\int dx = x + c$

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